

Stakeholder Analysis

1. Brand Managers (Business Users):

- *Need:* Track their company's standing relative to direct competitors.
- *System Expectation:* Access a straightforward, high-level dashboard showing sentiment trends and competitive rankings, backed by transparent sentiment scoring metrics.

2. Machine Learning & System Engineers (Technical Users):

- *Need:* Monitor system health, performance metrics, and data pipeline statuses.
- *System Expectation:* Access detailed model validation metrics (precision, recall, accuracy), input/output flow summaries, and error logs within a well-structured codebase.

3. End-Users / Consumers (General Users):

- *Need:* Easily compare different service providers before purchasing (e.g., choosing a telecom network or mobile wallet in Egypt).
- *System Expectation:* An intuitive, responsive interface with simple interactive visualizations and clear brand-to-brand comparisons.

User Stories

1. **User - Brand Manager:**

- “As a Telecom Brand Manager, I want to see our brand's positive feedback percentage alongside our direct competitors, so that I can measure the impact of our recent promotional campaign.”

2. **User - End Consumer:**

- “As a mobile wallet user in Egypt, I want to view the brand rankings for top fintech companies, so that I can choose a service provider with high user satisfaction and reliable support.”

3. **User - System Administrator:**

- “As an MLOps Engineer, I want to view the accuracy, precision, and recall metrics of our active classification model, so that I can easily spot and debug any potential model degradation issues.”

Functional Requirements (FR)

FR-1: Scraper Ingestion Engine

- **FR-1.1:** The system must dynamically generate query arrays in English and Arabic based on the target brand name.
- **FR-1.2:** The system must fetch posts and matching comments from the Reddit API.
- **FR-1.3:** The system must isolate Egyptian content using a specified list of localized keywords.

FR-2: Preprocessing and Normalization Engine

- **FR-2.1:** The system must strip HTML tags, URLs, diacritics, and punctuation from source text.
- **FR-2.2:** The system must preserve both English and Arabic characters during cleaning.
- **FR-2.3:** The system must apply rule-based stemming to normalize word variations before vectorization.

FR-3: Sentiment Classification

- **FR-3.1:** The system must convert clean text into numeric features using a 5000-feature TF-IDF vectorizer.
- **FR-3.2:** The system must categorize text into Positive, Neutral, or Negative classes using the LinearSVC model.
- **FR-3.3:** The system must save raw metrics and model predictions to structured local CSV datasets.

FR-4: Scoring & Competitive Ranking

- **FR-4.1:** The system must calculate brand sentiment scores using standard weighted variables.
- **FR-4.2:** The system must normalize user engagement based on post score and comment volume metrics.
- **FR-4.3:** The system must rank brands based on a weighted combination of sentiment, engagement, and sample size.

FR-5: Interactive Visualization Dashboard

- **FR-5.1:** The system must provide dropdown menus to let users choose specific brands to analyze.
- **FR-5.2:** The system must display comparative charts showing positive, neutral, and negative sentiment distribution.
- **FR-5.3:** The system must present a clear, interactive matrix showing final brand rankings and scores.

Non-Functional Requirements (NFR)

NFR-1: Execution & System Performance

- **NFR-1.1:** Local prediction on 100 queued records must complete in less than 5.0 seconds.
- **NFR-1.2:** The system must utilize caching protocols to keep dashboard load times under 2.0 seconds.

NFR-2: Security, Isolation, & Stability

- **NFR-2.1:** API credentials must be secured using .streamlit/secrets.toml or environmental variables.
- **NFR-2.2:** The scraping engine must handle network timeout failures gracefully without crashing.

NFR-3: Usability, Presentation, & Layout

- **NFR-3.1:** The dashboard must adapt cleanly to both mobile and desktop screens.
- **NFR-3.2:** Visualization charts must feature interactive tooltips and clear labels.

NFR-4: Reliability & Data Access

- **NFR-4.1:** The scraper must automatically fall back to global searches if regional filters return no results.
- **NFR-4.2:** The system must preserve historical predictions to prevent data loss when API connections fail.